EBM — Explainable Boosting Machine

What it is, how it works, and why it matters for this dissertation

Economic Crisis Prediction: EWS for Banking Crises — MSc AI Dissertation

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1. What EBM Is

EBM stands for Explainable Boosting Machine. It was developed by Lou, Caruana, Gehrke and Hooker (2013) at Microsoft Research. In this dissertation it is labelled **M2.5** and sits between Random Forest (M2) and XGBoost+FinBERT (M3) in the model sequence.

1.1 The Technical Definition

EBM is a **gradient-boosted Generalised Additive Model (GAM)**. A standard Generalised Additive Model predicts an outcome as the sum of individual functions of each feature:

$$\text{crisis risk} = f_1(\text{term\_spread}) + f_2(\text{credit\_growth}) + f_3(\text{house\_prices}) + \ldots + \text{intercept}$$

Each *f* is a **shape function** — a learned curve that maps a single feature value to its contribution to crisis probability in log-odds. The model's prediction is the sum of all these individual contributions.

Gradient boosting is the training algorithm — the same family used by XGBoost and Random Forest, but applied here to learn those shape functions one feature at a time, cycling through features repeatedly until convergence. This is what makes EBM more accurate than earlier GAMs: boosting allows it to capture complex non-linear patterns within each individual feature.

EBM also learns a small number of **pairwise interaction terms** (up to five in this dissertation's specification) — functions of two features jointly — which captures the most important cross-feature interactions without sacrificing interpretability.

2. What Makes EBM Different from RF and XGBoost

The critical distinction is in how feature contributions are computed:

Property	Random Forest (M2)	XGBoost (M3)	EBM (M2.5)
Model structure	Hundreds of deep trees	Hundreds of shallow trees	Sum of shape functions
Feature contributions	Tangled across trees	Tangled across trees	Exact, per-feature curves
Interpretability method	Needs SHAP (post-hoc)	Needs SHAP (post-hoc)	Shape functions are the model itself
Attribution accuracy	Approximate	Approximate	Exact
Regime instability detection	Indirect / confounded	Indirect / confounded	Direct — reads model parameters

For Random Forest and XGBoost, **SHAP is a separate approximation tool** applied after training. It estimates what each feature contributed, but those estimates can be wrong —

especially under distribution shift. For EBM, there is no approximation step. The shape function **is** the model's learned relationship. When you plot f(term_spread_lag3), you are reading the model's actual internal parameters directly, not an estimate of them.

3. Making Failure Visible Rather Than Merely Documenting It

The key phrase in the dissertation's contribution statement is that EBM **"make[s] the failure mechanism directly visible"** rather than merely documenting performance collapse. Here is precisely what that means.

3.1 The Regime Instability Test

In this dissertation, EBM is trained twice — once on a **pre-GFC window (2003–2006)** and once on a **GFC-inclusive window (2003–2010)** — and the shape functions for the same features are plotted side by side (Figure 4.7).

3.2 Why This Cannot Be Done with RF or XGBoost

If you performed the same comparison using SHAP values on a Random Forest, any differences across the two training windows would have **two possible explanations:**

1. The model genuinely learned a different feature-risk relationship
2. The SHAP approximation behaved differently under the two data distributions

You cannot separate these two explanations. The attribution tool introduces noise that is indistinguishable from signal.

For EBM, there is **only one explanation.** The shape functions are the model's exact fitted parameters. When the pre-GFC term_spread_lag3 shape function is monotonically increasing and the GFC-inclusive one reverses sign and crosses it — that is not an approximation artefact. That is a direct, unambiguous measurement of how the model's learned mapping changed between regimes.

> **CORE FINDING**
> At term_spread_lag3 — the highest-ranked EBM feature (native importance 1.717) — the same spread value is associated with reduced crisis risk in the pre-GFC regime and increased crisis risk in the GFC-inclusive regime. The curves cross. This sign reversal is read directly off the model's own parameters, not inferred from any post-hoc tool.

4. Why This Matters Beyond Performance Metrics

If this dissertation had only reported AUROC and AUPRC, the finding would be: *"EBM collapses to AUROC 0.333 under Split B."* That tells you the model failed. It does not tell you **why** or **through which mechanism.**

The shape function analysis converts the performance collapse into a legible explanation: the model trained on pre-GFC data learned that high three-year lagged term spreads signal elevated crisis risk. When evaluated on GFC data, where that relationship had reversed, it systematically assigned **low risk to high-risk observations** and **high risk to low-risk ones** — driving AUROC below random.

4.1 Parametric Confirmation

The panel logistic sign reversal provides independent confirmation from a completely different analytical framework: the lag-one term spread coefficient is $\beta = -0.646$ ($p = 0.004$) against lag-three $\beta = +0.630$ ($p = 0.003$). The logistic regression and the EBM shape function are pointing at the same structural property from different angles — together establishing that the term spread relationship is **non-monotonic across time** and cannot be captured by any model that treats it as fixed.

4.2 The Theoretical Implication

Because EBM shape functions are exact model parameters, the instability observed across training windows reflects a property of the **data-generating process** rather than an attribution approximation error. This places banking crisis prediction within the broader class of **non-stationary ML problems under distribution shift** — the dissertation's theoretical contribution (Section 1.4).

Static, pooled EWS models are therefore not merely empirically limited — they may be **conceptually misaligned** with structural break environments. A correctly specified model trained on one crisis regime will systematically misread the signal in the next. No amount of additional data, model complexity, or standard interpretability enhancements resolves this.

5. EBM's Role in the Model Sequence

EBM sits at a specific position in the dissertation's argument:

- **M1 (Logistic Regression)** — transparent, linear, constrained. Baseline.
- **M2 (Random Forest)** — non-linear, opaque. Better in-sample; still fails prospectively.
- **M2.5 (EBM)** — non-linear, interpretable. Higher in-sample metrics; worse prospective collapse (AUROC 0.333 vs RF 0.481). Makes the failure mechanism visible.
- **M3 (XGBoost + FinBERT)** — non-linear, opaque, sentiment-augmented. Tests whether text adds signal beyond macro.

EBM's worse prospective performance relative to RF is itself analytically important. It demonstrates that **the very flexibility that enables interpretability (shape functions) is the same mechanism that produces regime-specific overfitting**. This is not a conventional interpretability-performance trade-off — it is a unified structural explanation of why both interpretability and performance collapse simultaneously under distribution shift.

> **KEY POINT FOR THE DISSERTATION'S ARGUMENT**
> EBM should be adopted in EWS design for its interpretability contribution — the ability to make regime instability directly visible — not as a claim of prospective superiority. Its Split B AUROC of 0.333 is not a failure of the model; it is a measurement of the structural non-stationarity in the data-generating process, made legible by the model's native transparency.

Reference: Lou, Y., Caruana, R., Gehrke, J. and Hooker, G. (2013) 'Accurate intelligible models with pairwise interactions', *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 623–631.